Roya W. Parsa

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Education

Dartmouth College, Hanover, NH **Expected June 2027**
Master of Science, Computer Science **GPA 4.0/4.0**

- Honors/Awards: Guarini Graduate Merit Scholarship (75%)
- Relevant Coursework: Machine Learning, Computer Vision, AI Agents, Deep Learning, Computational Immunology
- Teaching Assistant: Graph-based ML, Machine Learning & Statistics

Adelphi University, Garden City, NY **May 2025**
Bachelor of Science, Major Computer Science and Minor in Statistics **GPA 4.0/4.0**

- Honors/Awards: Summa Cum Laude, Honors College, Dean's Circle, Honors College Summer Research Fellowship
- Relevant Coursework: Machine Learning, AI, Computational Biology, Software Engineering, Databases

Work Experience

Graduate Research Assistant (Hassanpour Lab) **Nov 2025 – Present**
Dartmouth College, Hanover, NH

- Developed a multimodal classification framework integrating Prov-GigaPath whole-slide pathology embeddings with clinical serology to distinguish 4 liver diseases, achieving 73% held-out accuracy and 0.72 macro-F1 across 267 slides from 197 patients
- Investigated representation-level failure modes in medical foundation models, identifying substantial AIH-to-PBC (31.5%) and PSC-to-PBC (29.8%) misclassification, revealing challenges in distinguishing clinically related disease subtypes
- Co-leading development of a real-time YOLOv8 ultrasound detection system for cervical cancer screening, optimizing computer vision inference for deployment in resource-constrained clinical environments
- Implemented Low-Rank Adaptation (LoRA) for parameter-efficient fine-tuning of the Prov-GigaPath slide encoder, adapting pre-trained pathology representations to 4-class liver disease classification while reducing trainable parameters compared to full-model fine-tuning
- Investigating temporal distribution shift and recursive synthetic-data generation in medical foundation models for MS thesis, designing controlled experiments across 5 drift scenarios to quantify performance degradation and evaluate mitigation strategies
- Engineered a reproducible HPC pipeline for 1,200+ gigapixel pathology images using SLURM and Apptainer, automating tissue segmentation, patch extraction, stain normalization, and feature generation to reduce preprocessing time from days to hours

Student Researcher (PI: Dr. Dong Han) **Mar 2024 – Nov 2025**
New York Proton Center, New York, NY

- Developed SPR-NET, a full-stack ML application for automated neural network training and stopping-power prediction; a custom PyTorch U-Net improved performance over analytical methods by 60% in high-density regions
- Built Python benchmarking pipelines comparing dual-energy and single-energy CT methods to evaluate robustness under imaging artifacts and tissue differentiation challenges; manuscript in review
- Developed a clinical CT analysis suite generating stopping-power maps with sub-1% SPR deviation relative to Gammex-provided reference SPR maps, supporting quantitative evaluation of proton therapy planning methods

Software Engineer Intern **Jun 2023 – Dec 2024**
North Atlantic Industries, Bohemia, NY

- Built a GPT-powered documentation search system integrating semantic retrieval and natural-language query processing to improve access to technical information, reducing support tickets by 50%
- Engineered a Python ETL pipeline processing 8,000+ unstructured technical documents into 25+ relational SQL tables weekly, keeping downstream engineering data current within one business day
- Containerized VxWorks cross-compilation toolchains using Docker and deployed build images on AWS EC2 to improve build reproducibility across Linux environments

Selected Projects

Fairness Audit of Shortcut Mitigation Across Skin Types GitHub

- - Developed a pre-training bias detection framework for dermatology AI, evaluating 6 dataset-level signals across 3,752 images and 20 simulated curation scenarios; identified visual-distribution metrics that predicted demographic performance disparities (AUROC 0.83), outperforming traditional class-imbalance measures (0.60).

Bitty: Event-Drived AI Desktop Agent GitHub

- Engineered an event-driven desktop AI agent with asynchronous LLM inference, extensible tool execution, and permission-gated actions; integrated real-time animated UI states with agent lifecycle events to deliver transparent, responsive AI interactions without additional inference overhead

SimpleCoder: LLM Coding Agent with RAG GitHub

- Developed a Python coding agent integrating repository-grounded RAG, modular planning, context management, and permission controls; implemented CLI and Streamlit interfaces for interactive code-generation workflows

Awards & Fellowships

- **AI Research Fellowship, The Ohio State University (2026–2027):** Fully funded, competitive year-long fellowship in bioinformatics and imageomics on site in Hawaii; Investigating computational solutions for local flora degradation
- **Honors College Summer Research Fellowship, Adelphi University (2024):** $4,000 award supporting undergraduate research in machine learning for proton therapy

Skills

Data & Infrastructure: Python, SQL, ETL, large-scale data preprocessing, SLURM, Apptainer, Docker, AWS, Linux, Git, Jenkins
ML/AI: PyTorch, Hugging Face, TensorFlow, scikit-learn, foundation models, VLMs, YOLO, OpenCV, LangChain, RAG
Evaluation & Other: Experimental design, model benchmarking, PCA/UMAP, R, MATLAB, JavaScript, NumPy, Pandas, SciPy, FastAPI, Flask